Smart Helmet for Accident Prevention and Safety

A Framework for Multi-Objective Optimization and Dynamic Scheduling using Microservices.

Report Submitted to the Garden City University
as the requirement for the course

10ABTEC22758: PILOT PROJECT

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Bonafide Certificate

This is to certify that the report titled **"Smart Helmet for Accident prevention And Safety"** submitted in partial .fulfilment of the requirements for the award of the degree of B. Tech. Computer Science & Engineering (Data Science) to the Garden City University, is a bonafide record of the work done by S Shalini (22BTDS174) ,NavaneethYadav (22BTDS119),Thanushree N S (22BTDS155) and Anushree V S (22BTDS145) during the 7th semester of the academic year 2025-26, in the Dr. APJ Abdul Kalam School of Engineering, under my supervision. This report has not formed the basis for the award of any degree, diploma, associateship, fellowship or other similar title to any candidate of any University.

Signature of the Pilot Project Co-ordinator :

Name and Affiliation :

Date :

Review held on _____

Examiner 1 **Examiner 2**

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Declaration

We declare that the report titled **"Smart Helmet for Accident prevention And Safety"** submitted by us is an original work done by us under the guidance of **Prof Mr. Jhon, Dept. of Engineering** during the7th semester of the academic year 2025-26, in the **Dr. APJ Abdul Kalam School Engineering**. The work is original and wherever we have used materials from other sources, we have given due credit and cited them in the text of the thesis. This has not formed the basis for the award of any degree, diploma, associate-ship, fellowship or other similar title to any candidate of any University

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Abstract

The current generation of road-safety devices is largely reactive, activating protections only after an accident has already occurred, which significantly limits their ability to prevent fatalities or reduce injury severity. This pilot project addresses this limitation by developing an intelligent Smart Helmet designed for proactive accident prevention and real-time safety monitoring. The core objective is to transform the traditional helmet from a passive protective gear into an active, anticipatory safety system. The Smart Helmet integrates advanced sensing technologies such as alcohol detection, impact detection, and helmet-wear verification, supported by an embedded micro-controller for continuous monitoring. The system leverages wireless communication and IoT - based data transmission to detect critical risk events (e.g., collisions, loss of balance, unsafe riding conditions) and automatically trigger emergency protocols. Once activated, the system proactively sends real-time alerts, including GPS location, to designated emergency contacts to ensure timely assistance. Its technical foundation rests on a modular hardware-software architecture to ensure reliability and scalability. Central to its intelligence is a sensor-fusion algorithm that interprets multi-sensor inputs to accurately infer rider safety conditions. Accident prediction and emergency response optimization are achieved using threshold-based decision models and multi-parameter evaluation techniques, enabling efficient resolution of competing safety factors such as response time, accuracy, and environmental noise.

Project Documentation:

Smart Helmet for Accident Prevention and Safety

1. Introduction

The Smart Helmet for Accident Prevention and Safety represents a pivotal advancement in modern road-safety technology, transforming the traditional helmet from a passive protective device into an intelligent, anticipatory safety system. In today's world, riding two - wheelers involves navigating unpredictability—rash driving, poor road conditions, alcohol-influenced riding, and delayed emergency response contribute significantly to accident severity. The Smart Helmet bridges these gaps by becoming a central, proactive guardian of rider safety.

By integrating real-time sensors such as alcohol detection, impact detection, helmet-wear verification, and GPS tracking, the Smart Helmet gains continuous situational awareness of the rider's safety. Whether it's detecting an unsafe alcohol level, identifying whether the rider is wearing the helmet correctly, or sensing a high-impact collision, the system intelligently interprets these conditions and takes protective action.

This feature leverages IoT-based communication, real-time monitoring, intelligent rule-based decision models, and multi-parameter risk analysis to deliver safety responses that feel immediate and life-saving. The Smart Helmet doesn't simply detect events—it predicts risk, prevents unsafe riding, and ensures help arrives when needed. The result is a system that quietly works in the background, converting raw sensor data into actionable safety decisions. The Smart Helmet is not merely an enhancement; it is a redefinition of how rider safety can be anticipated, monitored, and ensured.

New Feature: Proactive Itinerary Intelligence

2. Feature Overview

The Intelligent Safety Detection System elevates the Smart Helmet from a basic protective device to a predictive, IoT-enabled safety companion. By integrating sensors and wireless communication modules, the Smart Helmet automatically detects unsafe conditions such as alcohol consumption, helmet absence, collisions, and abnormal riding patterns. Instead of waiting for rider input, the system proactively prevents ignition, alerts emergency contacts, and monitors rider safety throughout the ride.

This feature shifts the Smart Helmet’s role from simply "protecting during an accident" to "actively preventing accidents and ensuring rapid emergency response."

3. Core Capabilities

1. Helmet Wearing Detection

A pressure or IR-based sensor ensures the helmet is worn.

Ignition will not start unless the helmet is detected.

2. Alcohol Detection

The MQ-3 sensor checks the rider’s breath alcohol level.

If alcohol exceeds a defined limit → **Ignition remains locked**.

3. Accident Detection

A vibration/accelerometer sensor detects sudden impact.

If detected → System triggers emergency protocol.

4. Real-Time GPS Tracking

Location is sent automatically to emergency contacts.

5. Emergency Alert System

Upon accident detection, the system sends:

1. SMS to emergency contacts
2. GPS coordinates
3. Auto-call (optional via GSM module)

6. Bike Ignition Control

Helmet + No alcohol + No abnormal sensor readings → Only then ignition is enabled.

4. User Scenario: "The Rider Safety Sequence "

Step 1: Start of Journey

Rider wears the smart helmet → Helmet sensor activates → Bike ignition unlocks.

Step 2: Alcohol Safety Check

Helmet's MQ-3 sensor scans breath.
If alcohol detected → Bike stays locked and alert is displayed.

Step 3: Riding Phase

All sensors remain active, monitoring:

1. Speed
2. Tilt
3. Vibration
4. Unusual impact

Step 4: Accident Occurs

Impact sensor triggers → Microcontroller identifies crash → GPS coordinates are retrieved.

Step 5: Automatic Alert

System sends emergency SMS:

> "Accident detected. Rider location: <GPS Link>"

Response team and family can immediately locate the rider.

5. Proposed Technical Implementation

- **Hardware Components :**
  - Microcontroller (Arduino / NodeMCU / ESP32)
  - MQ-3 alcohol sensor
  - Accelerometer/gyro (MPU6050)
  - IR/pressure detection sensors
  - GSM/GPRS module (SIM800L)
  - GPS module (NEO-6M)
  - Ignition control relay
  - Lithium Battery.
- **Software Architecture :**
  - Embedded C / Python
  - IoT communication (MQTT)
  - Cloud storage (optional)
  - Mobile app alerts
- **Working principle :**
  - Helmet module continuously sends data to bike module via RF/WiFi.
  - Ignition module unlocks only under safe conditions.
  - Crash detection triggers emergency protocol.
  - GPS data is fetched and sent via GSM module.

- System logs data for post-analysis.

6. Key User Benefits

- **Accident Prevention :** Prevents drunk driving accidents.
- **Alcohol-Free Riding :**Automatic SOS reduces fatalities caused by delayed help.
- **Faster Emergency Response :** Ensures helmet is always worn..
- **Low cost Implementation:** Uses affordable sensors with high reliability.

7. Conclusion

The Smart Helmet for Accident Prevention and Safety elevates traditional road-safety gear into an intelligent, IoT-powered protective system. With its multi-sensor awareness, predictive alerts, automated emergency responses, and real-time monitoring, the Smart Helmet transcends conventional protective behavior by taking initiative—detecting unsafe conditions, preventing high-risk actions, and responding rapidly during emergencies. Every safety decision the helmet makes reflects a deep understanding of rider behavior, environmental risk, and critical accident dynamics. Ultimately, this intelligent safety system transforms the helmet into a dynamic, context-aware companion capable of managing complexity with precision. It reinforces the promise of delivering a safe, seamless riding experience—one where the rider focuses on the journey while the Smart Helmet continuously ensures their protection. This project sets a new standard for intelligent safety systems, positioning the Smart Helmet as a transformative solution in accident prevention and rider protection.